Introduction to Machine Learning

What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. Instead of writing rules manually, ML systems identify patterns from data and make decisions with minimal human intervention. The term was coined by Arthur Samuel in 1959 during his work on a checkers-playing program at IBM.

Types of Machine Learning

There are three main types of machine learning. Supervised learning uses labeled training data to learn a mapping from inputs to outputs. Common examples include spam detection and image classification. Unsupervised learning finds hidden patterns in data without labels - clustering and dimensionality reduction fall in this category. Reinforcement learning trains an agent to make decisions by rewarding desired behaviors and penalizing undesired ones, commonly used in game-playing AI and robotics.

Neural Networks and Deep Learning

Neural networks are computational models inspired by the human brain. They consist of layers of interconnected nodes called neurons. Deep learning refers to neural networks with many hidden layers, capable of learning complex representations. Convolutional Neural Networks (CNNs) excel at image tasks, while Recurrent Neural Networks (RNNs) handle sequential data like text. Transformers, introduced in 2017, revolutionized natural language processing through the self-attention mechanism.

Large Language Models (LLMs)

Large Language Models are deep learning models trained on massive text corpora to understand and generate human language. GPT (Generative Pre-trained Transformer) by OpenAI and Claude by Anthropic are prominent examples. These models use the transformer architecture and are pre-trained on billions of tokens. LLMs can perform tasks like summarization, translation, question answering, and code generation through a technique called prompting.

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is a technique that enhances LLM responses by retrieving relevant documents from an external knowledge base at query time. Instead of relying solely on parametric knowledge baked into model weights, RAG fetches context from a vector store and feeds it to the LLM. The pipeline typically involves chunking documents, converting them to embeddings using an embedding model, storing them in a vector database like FAISS or Chroma, and then at inference time retrieving the top-k most similar chunks and passing them as context to the LLM. This approach reduces hallucinations and keeps the model grounded in factual source documents.

Vector Databases and Embeddings

Embeddings are dense numerical representations of text that capture semantic meaning. Similar texts have vectors close together in embedding space. Popular embedding models include sentence-transformers (e.g., all-MiniLM-L6-v2) and OpenAI's text-embedding-ada-002. Vector databases like FAISS, Chroma, Pinecone, and Weaviate are optimized for fast similarity search over millions of vectors using algorithms like HNSW and IVF. FAISS (Facebook AI Similarity Search) is a popular open-source library that runs locally, making it ideal for experimentation and learning.

Applications of Machine Learning

Machine learning powers many real-world applications. In healthcare, ML models assist in disease diagnosis from medical images and drug discovery. In finance, they detect fraudulent transactions and power algorithmic trading. Natural language processing enables virtual assistants, sentiment analysis, and machine translation. Computer vision is used in self-driving cars, surveillance, and quality control in manufacturing. Recommendation systems on platforms like Netflix and Spotify use collaborative filtering and content-based methods to personalize content.